

Doob's Proof of MLE Consistency

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Abstract

We present a concise exposition of Doob's proof of MLE consistency using surprisal and relative information. We clarify historical entropy terminology and explicitly verify all assumptions for the multivariate normal distribution using moment-generating functions and elementary matrix algebra.

Keywords: maximum likelihood estimator, true parameter, surprisal, relative information

1 Introduction

The consistency of the maximum likelihood estimator (MLE) of [10] is a basic result discovered more than a century ago. This method is typically treated as a special case of M -estimators, and derived via standard techniques dating back to [6] and [25]. A modern treatment is [24]. Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found the [9] consistency proof of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter θ spaces and general state X spaces with no extra effort only adds to its appeal.

The contribution of this note is expository: it gives a formulation of Doob's consistency argument in surprisal/information terminology, emphasizing the role of properness and separability, and works out the multivariate normal example in detail. The proofs are designed to be self-contained throughout: in particular, the consistency argument and the matrix lemmas are developed directly, without relying on determinant multiplicativity, eigenvalue continuity, or other standard results that can obscure the elementary structure of the argument.

We start with the setup and assumptions, then state and prove consistency. After that, we give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

2 Surprisal and Information

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X . We assume the density $p(X, \theta^*)$ of X is one of a parametric family $p(X, \theta)$ of densities, and we wish to recover θ^* from the samples $X_1, X_2, \dots$.

Stated in this manner, all we have to work with are the true expectation E_* and the family $p(X, \theta)$. While $p(X, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over or knowledge of E_* . Given this, how do we use the samples $X_1, X_2, \dots$ to recover θ^* ?

Define the *surprisal* [18, 22]

$$S(X, \theta) = \log(1/p(X, \theta)).$$

This terminology is intuitive: low probability equals high surprisal. Then it is natural to characterize θ^* as a minimizer of the expected surprisal $E_*(S(X, \theta))$. More precisely, we say θ^* is the *true parameter* if $S(X, \theta)$ is integrable for all θ and θ^* is the *unique* minimizer of $E_*(S(X, \theta))$. In particular, if θ^* is the true parameter

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and $\theta \neq \theta^*$, then $p(X, \theta) \neq p(X, \theta^*)$. This last property is usually stated separately as *identifiability*. Here for simplicity it is incorporated into the definition of true parameter.

Equivalently, given θ^* , the *relative information* is

$$I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)). \quad (1)$$

The relative information is the excess expected surprisal of θ over θ^* . Then θ^* is the true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ , with $I(\theta^*, \theta) = 0$ only if $\theta = \theta^*$. The relative information I appears in many fields, in different forms. Because of widely differing terminology, below we give an account of its history.

A special case is when there is a background expectation E such that

$$E_\theta(f(X)) = E(p(X, \theta)f(X))$$

is an expectation for all θ . While $E_\theta(1) = 1$, here E need not satisfy $E(1) = 1$, $E(1) = \infty$ is allowed. When this holds and $E_{\theta^*} = E_*$, the relative information takes the familiar form

$$I(\theta^*, \theta) = E \left(p(X, \theta^*) \log \left(\frac{p(X, \theta^*)}{p(X, \theta)} \right) \right), \quad (2)$$

$I(\theta^*, \theta) \geq 0$ for all θ , and $I(\theta^*, \theta) = 0$ implies $p(X, \theta) = p(X, \theta^*)$ [8, 7].

The simplest example is as follows. A vector $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$ is a probability vector and a random variable X satisfies $P_*(X = i) = \theta_i^*$, $i = 1, 2, \dots, d$. Then the mass function of X is $p(X, \theta^*) = \theta_X^*$. If θ is any other probability vector, the surprisal is $S(X, \theta) = \log(1/\theta_X)$, and

$$I(\theta^*, \theta) = \sum_{i=1}^d \theta_i^* \log(\theta_i^*/\theta_i). \quad (3)$$

Turning back to the general setting, the *likelihood* and (empirical) *mean surprisal*¹ are

$$L_n(\theta) = \prod_{k=1}^n p(X_k, \theta), \quad \bar{S}_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^n S(X_k, \theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes $E_*(S(X, \theta))$. By the (strong) law of large numbers (LLN), $\bar{S}_n(\theta)$ converges to $E_*(S(X, \theta))$ as $n \rightarrow \infty$, with probability one. Hence it is natural to estimate θ^* by a minimizer of $\bar{S}_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ .

Under **A** and **B** below, with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n . In what follows, only E_* and $p(X, \theta)$ enter, E and E_θ play no role.

3 Consistency

In addition to the existence of the true parameter, we make two assumptions. A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c . Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the first assumption

A. (Properness): There is an $N \geq 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,

$$F(\theta) \leq \bar{S}_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N. \quad (4)$$

We will need the LLN convergence $\bar{S}_n(\theta) \rightarrow E_*(S(X, \theta))$ across θ , in the sense

$$\bar{S}_n(\theta) \rightarrow E_*(S(X, \theta)), \quad \text{as } n \rightarrow \infty, \quad \text{for all } \theta, \quad (5)$$

¹Terminology analogous to mean loss in machine learning.

with probability one. For this we need the second assumption

B. (*Continuity*): The parameter space is separable and there is a continuous $\delta(\theta, \theta')$ and an integrable nonnegative $g(X)$ satisfying

$$|S(X, \theta) - S(X, \theta')| \leq \delta(\theta, \theta')g(X) \quad \text{and} \quad \delta(\theta, \theta) = 0. \quad (6)$$

Under **B**, a standard separability argument, given in the appendix, establishes (5) with probability one. Under **B**, the mean surprisal is continuous in θ , hence sublevel sets are closed. Under **A**, every sublevel set of the mean surprisal is contained in a compact sublevel set of $F(\theta)$, and hence is compact. Thus, under **A** and **B**, the mean surprisal is proper. Since a proper continuous function has a minimizer, an MLE $\hat{\theta}_n$ exists for $n \geq N$, with probability one.

A and **B** are particularly tractable for *exponential families* [5, 15, 19, 24], the densities $p(X, \theta)$ where $S(X, \theta)$ is a sum of products of the form $f(\theta)g(X)$.

In (5), there is no presumption of uniform convergence. This is a key feature of Doob's proof: no uniform convergence of the mean surprisal is required. The argument instead combines properness, a point-wise LLN, and the continuity estimate in (6) to control limit points of the MLE sequence.

Theorem (Consistency). *Let $X_1, X_2, \dots$ be independent and identically distributed, and let $p(X, \theta)$ be a parametric family with true parameter θ^* . Under **A** and **B**, with probability one, an MLE sequence $\hat{\theta}_n$ may be defined for sufficiently large n , and, for any MLE sequence, $\hat{\theta}_n$ converges to θ^* as $n \rightarrow \infty$.*

Proof. By the LLN,

$$\frac{1}{n} \sum_{k=1}^n g(X_k) \rightarrow E_*(g(X)), \quad \text{as } n \rightarrow \infty, \quad (7)$$

with probability one. Fix samples for which (4), (5) and (7) hold. We show $\hat{\theta}_n \rightarrow \theta^*$ for these samples. By (4),

$$F(\hat{\theta}_n) \leq \bar{S}_n(\hat{\theta}_n) \leq \bar{S}_n(\theta^*), \quad \text{for } n \geq N.$$

Since $\bar{S}_n(\theta^*) \rightarrow E_*(S(X, \theta^*))$, the right side is bounded. Since $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n \rightarrow \theta$ along a subsequence. By (6),

$$\bar{S}_n(\theta) - \bar{S}_n(\theta^*) \leq \bar{S}_n(\theta) - \bar{S}_n(\hat{\theta}_n) \leq \delta(\theta, \hat{\theta}_n) \cdot \frac{1}{n} \sum_{k=1}^n g(X_k). \quad (8)$$

Since the left side of (7) is bounded as $n \rightarrow \infty$, and $\delta(\theta, \hat{\theta}_n) \rightarrow 0$ along a subsequence, the right side of (8) goes to zero along a subsequence. By (1) and (5), the left side of (8) converges to $I(\theta^*, \theta)$, hence $I(\theta^*, \theta) \leq 0$. But $I(\theta^*, \theta) \geq 0$, hence $I(\theta^*, \theta) = 0$, showing $\theta = \theta^*$. Thus $\hat{\theta}_n \rightarrow \theta^*$ as $n \rightarrow \infty$. $\square$

4 Terminology History

The relative information $I(\theta^*, \theta)$ is often called “relative entropy” [5, 15]. The term “entropy,” originating in thermodynamics, was coined by [4]. [2] derived his H theorem, using a quantity H proportional to the negative of entropy. [11], building on the combinatorial interpretation of entropy [3, 17], reversed Boltzmann's sign convention, writing entropy with the sign convention used today.

[20] independently defined

$$H(p) = - \sum_{i=1}^d p_i \log p_i.$$

[16] had earlier defined quantum entropy. According to a widely repeated account attributed to Shannon and reported by [23], von Neumann suggested the name “entropy” when Shannon was casting about for terminology after developing his communication theory in the mid-1940s; [21] later expressed doubt that the

conversation took place. Whatever the origin of the name, in his 1948 paper, Shannon explicitly links his H to Boltzmann's: “ H is then, for example, the H in Boltzmann's H theorem,” connecting his terminology to the earlier usage despite the difference in sign conventions.

The relative information $I(\theta^*, \theta)$ (2) is jointly convex in the densities $p(X, \theta^*)$ and $p(X, \theta)$, as is apparent in the special case (3), whereas the entropy $H(p)$ above is strictly concave in p . From this perspective, the terminology “relative entropy” can seem counterintuitive. This terminology extends into the Python `scipy.stats` package: `entropy(p)` returns the concave H , whereas `entropy(p,q)` returns the convex I .

This situation can be understood by remembering the conceptual opposition between entropy and information: entropy is concave, whereas information is convex. The two notions are closely related, but differ by sign and by the role they play in optimization.

Indeed, based on curvature, one might argue that the *negative* of (3) is the more natural candidate for “relative entropy.” Similar remarks apply to the “cross-entropy” loss J of machine learning. Since J differs from I by a constant, one might analogously call J the *cross-information*, and its negative “cross-entropy.”

While many authors use terminology different from ours, some align partially with our usage. [19] uses “Kullback-Leibler information” for I , which is more descriptive than the widely used “Kullback-Leibler divergence” [1, 14]. In particular, [13] refers to I as an “information function.” [8] denote their rate function by I rather than H ; the relative information is the i.i.d. special case of their I .

Doob's 1934 proof appeared at the dawn of modern probability theory, within the milieu of Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, and Wiener, and within a year of the [12] extension theorem.

5 Multivariate Normal

Let μ be in $\mathbf{R}^d$, Q a positive symmetric $d \times d$ matrix, and $\theta = (\mu, Q)$. Let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With X in $\mathbf{R}^d$, the *normal density with mean μ and variance Q* is $p(X, \theta) = e^{-S(X, \theta)}$ where

$$S(X, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} (X - \mu) \cdot Q^{-1} (X - \mu). \quad (9)$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter $\theta^* = (\mu^*, Q^*)$* if its moment-generating function is

$$E_*(e^{v \cdot X}) = \exp \left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v \right). \quad (10)$$

Suppose X is normally distributed with parameter $\theta^* = (\mu^*, Q^*)$ as in (10), and let $X_1, X_2, \dots$ be independent and identically distributed copies² of X . We show θ^* is the true parameter, and we verify **A** and **B**.

Assumption **B** follows from (9). To establish **A**, we review basic matrix facts, stated as lemmas and derived in the appendix. The proofs deliberately avoid invoking determinant multiplicativity or continuity properties of eigenvalues: the point is not merely to verify the required facts, but to expose the elementary mechanism behind them.

We say a symmetric matrix Q is *positive*, and we write $Q > 0$, if $v \cdot Qv > 0$ for $v \neq 0$. We say a symmetric matrix Q is *nonnegative*, and we write $Q \geq 0$, if $v \cdot Qv \geq 0$ for all v . Given symmetric matrices Q, R we write $Q \geq R$ if $Q - R \geq 0$. If $Q > 0$, then $Q \geq \delta I$ for some $\delta > 0$. If $Q \geq 0$ and $R \geq S$, then $\text{trace}(QR) \geq \text{trace}(QS)$. Moreover $\text{trace}(QR) = \text{trace}(RQ)$ and we have the Cauchy-Schwarz inequality

$$\text{trace}(ZR) \leq \sqrt{\text{trace}(Z^2)} \sqrt{\text{trace}(R^2)}, \quad Z, R \text{ symmetric.}$$

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v \otimes w$ is the matrix whose ij -th entry is $v_i w_j$. Then $v \otimes v$ is nonnegative and

$$v \cdot Qv = \text{trace}(Q(v \otimes v)).$$

Let $R = v \otimes v$. Since $R^2 = |v|^2 R$ and $\text{trace}(R) = |v|^2$, $\text{trace}(R^2) = |v|^4$. Using this and Cauchy-Schwarz,

$$\delta^2 = \text{trace}(Z^2) \implies Z \leq \delta I. \quad (11)$$

²This setup is a special case of the Kolmogorov extension theorem.

If $u = \sqrt{\epsilon}v - w/\sqrt{\epsilon}$, then $u \otimes u \geq 0$, hence

$$v \otimes w + w \otimes v \leq \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w. \quad (12)$$

If Z_n is a sequence of symmetric matrices with $Z_n \rightarrow 0$, then $\text{trace}(Z_n^2) \rightarrow 0$, hence by (11), given $Q > 0$, there is $N \geq 1$ such that $Z_n \leq Q$ for $n \geq N$.

By the eigenvalue decomposition, every nonnegative matrix is a sum of matrices of the form $v \otimes v$, and every positive matrix is a sum of δI and matrices of the form $v \otimes v$.

The determinant $\det(Q)$ is the product of its eigenvalues.

Lemma 1. *With Q invertible symmetric and t a scalar, $\det(Q + tv \otimes v) = \det(Q) (1 + tv \cdot Q^{-1}v)$.*

Lemma 2. *If $S > 0$ and $Z > 0$, then $\det(SZS) = \det(Z) \det(S)^2$.*

Lemma 3. *For c scalar, let $\epsilon = e^{-c}$ and $h(\lambda) = \log \lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$, and $h(\lambda) \leq c$ implies $\epsilon \leq \lambda \leq 1/\epsilon$.*

Lemma 4. *For $Q^* > 0$ and c scalar, let $\epsilon = e^{-2c-1} \det(2\pi e Q^*)$ and*

$$F_*(Q, Q^*) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1} Q^*), \quad Q > 0. \quad (13)$$

Then $F_(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \leq c$ implies $\epsilon Q^* \leq Q \leq Q^*/\epsilon$.*

Let $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$. Given $\theta = (\mu, Q)$, let

$$F(\mu, Q) = F_*(Q, f(\mu)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

By Lemma 4, (μ^*, Q^*) is the unique minimizer of $F(\mu, Q)$.

In (10), multiply both sides by $e^{-v \cdot \mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$E_*(X) = \mu^*, \quad E_*((X - \mu) \otimes (X - \mu)) = f(\mu).$$

It follows $E_*(S(X, \theta)) = F(\mu, Q)$, and $\theta^* = (\mu^*, Q^*)$ is the true parameter.

The second consequence of Lemma 4 is properness of $F(\mu, Q)$. If $F(\mu, Q) \leq c$, then $F_*(Q, Q^*) \leq c$, hence $\epsilon Q^* \leq Q \leq Q^*/\epsilon$, confining Q to a compact subset of the space of positive definite matrices. From this,

$$\begin{aligned} c &\geq F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu) \\ &\geq F_*(Q^*, Q^*) + \frac{\epsilon}{2}(\mu^* - \mu) \cdot (Q^*)^{-1}(\mu^* - \mu), \end{aligned}$$

confining μ to a compact subset of $\mathbf{R}^d$. Thus $F(\mu, Q)$ is proper.

With

$$Q_n(\mu) = \frac{1}{n} \sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu), \quad \hat{\mu}_n = \frac{1}{n} \sum_{k=1}^n X_k, \quad \hat{Q}_n = Q_n(\hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$, and let $0 < \epsilon < 1/2$.

By the LLN, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \rightarrow 0, \quad Q^* - Q_n(\mu^*) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

Thus there is $N \geq 1$, depending on the samples, such that, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \leq \epsilon^2 Q^*, \quad Q_n(\mu^*) \geq (1 - \epsilon) Q^*, \quad \text{for } n \geq N.$$

Let

$$f_\epsilon(\mu) = (1 - \epsilon)f(\mu) - \epsilon Q^*, \quad F_\epsilon(\theta) = F_\epsilon(\mu, Q) = F_*(Q, f_\epsilon(\mu)).$$

Since $f_\epsilon(\mu) \geq (1 - 2\epsilon)f(\mu)$,

$$F_\epsilon(\mu, (1 - 2\epsilon)Q) \geq \frac{1}{2} \log \det((1 - 2\epsilon)I) + F(\mu, Q),$$

hence $F_\epsilon(\theta)$ is proper. With $w = \hat{\mu}_n - \mu^*$ and $v = \mu - \mu^*$, using (12),

$$v \otimes w + w \otimes v \leq \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \leq \epsilon f(\mu)$$

for all μ and $n \geq N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$Q_n(\mu) = Q_n(\mu^*) - (v \otimes w + w \otimes v) + v \otimes v.$$

Combined with the above inequalities, this leads to

$$Q_n(\mu) \geq f_\epsilon(\mu), \quad \text{for all } \mu \text{ and } n \geq N.$$

In particular, since $f_\epsilon(\mu) \geq (1 - 2\epsilon)Q^*$, we have shown $\hat{Q}_n > 0$ for $n \geq N$.

Since $\bar{S}_n(\theta) = F_*(Q, Q_n(\mu))$ and $Q_n(\mu) \geq \hat{Q}_n$ by least squares, by Lemma 4,

$$\bar{S}_n(\theta) = F_*(Q, Q_n(\mu)) \geq F_*(Q, \hat{Q}_n) \geq F_*(\hat{Q}_n, \hat{Q}_n) = \bar{S}_n(\hat{\theta}_n),$$

for all θ and $n \geq N$. Thus $\hat{\theta}_n$ is the MLE for $n \geq N$, and

$$\bar{S}_n(\theta) = F_*(Q, Q_n(\mu)) \geq F_\epsilon(\theta), \quad \text{for all } \theta \text{ and } n \geq N,$$

establishing [A](#).

6 Disclosures

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A Auxiliary Proofs

Proof of (5) with probability one. By separability of the parameter space, there is a countable dense set D of parameters. Since D is countable, with probability one,

$$\bar{S}_n(\theta') \rightarrow E_*(S(X, \theta')), \quad \text{as } n \rightarrow \infty, \quad \text{for all } \theta' \text{ in } D. \quad (14)$$

Fix samples for which (7) and (14) hold. We establish (5) for these samples. If θ, θ' are any parameters, by (6) and the triangle inequality,

$$\begin{aligned} |\bar{S}_n(\theta) - E_*(S(X, \theta))| &\leq |\bar{S}_n(\theta') - E_*(S(X, \theta'))| \\ &\quad + \delta(\theta, \theta') \left(E_*(g(X)) + \frac{1}{n} \sum_{k=1}^n g(X_k) \right). \end{aligned} \quad (15)$$

Now take the $\limsup_{n \rightarrow \infty}$ in (15). By (7) and (14),

$$\limsup_{n \rightarrow \infty} |\bar{S}_n(\theta) - E_*(S(X, \theta))| \leq 2\delta(\theta, \theta') E_*(g(X)), \quad \text{for all } \theta' \text{ in } D \text{ and all } \theta.$$

Since D is dense, $\delta(\theta, \theta')$ is continuous, and $\delta(\theta, \theta) = 0$, for fixed θ and varying θ' , the right side is arbitrarily small, while the left side does not depend on θ' . This establishes (5). $\square$

Proof of Lemma 1. Assume $t \neq 0$. If v is orthogonal to an eigenvector x of Q , then we reduce to a lower-dimensional case by restricting to $x^\perp$. Hence we may assume $v \cdot x \neq 0$ for all eigenvectors x of Q . If v is orthogonal to an eigenvector x' of $Q' = Q + tv \otimes v$, then x' is an eigenvector of Q , contradicting our assumption. Hence we may assume $v \cdot x' \neq 0$ for all eigenvectors x' of Q' . If x'_1 and x'_2 are linearly independent eigenvectors of Q' corresponding to the same eigenvalue, then, for suitable r_1, r_2 , $x' = r_1 x'_1 + r_2 x'_2$ is an eigenvector of Q' satisfying $v \cdot x' = 0$, contradicting our assumptions. Hence we may assume the eigenvalues of Q' are simple. If x and x' are eigenvectors for Q and Q' corresponding to the same eigenvalue, then

$$t(v \cdot x')(v \cdot x) = x' \cdot (tv \otimes v)x = x' \cdot Q'x - x' \cdot Qx = 0,$$

contradicting our assumptions. Hence we may assume the eigenvalues of Q and the eigenvalues of Q' do not overlap. Let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Q and $\lambda'_1, \lambda'_2, \dots, \lambda'_d$ the eigenvalues of Q' . Let $v_1, v_2, \dots, v_d$ be the corresponding orthonormal basis of eigenvectors of Q . Then the *secular equation* is

$$\sigma(z) = 1 + tv \cdot (Q - zI)^{-1}v = 1 + t \sum_{i=1}^d \frac{(v \cdot v_i)^2}{\lambda_i - z} = 0.$$

If λ' is an eigenvalue of Q' with eigenvector x' , then $(Q' - \lambda'I)x' = 0$, hence $(Q - \lambda'I)x' + t(v \cdot x')v = 0$. Taking the dot product with $(Q - \lambda'I)^{-1}v$ yields $v \cdot x' + t(v \cdot x')(v \cdot (Q - \lambda'I)^{-1}v)$, hence $\sigma(\lambda') = 0$. If we let

$$\det(Q - zI) = \prod_{i=1}^d (\lambda_i - z), \quad p(z) = \det(Q - zI)\sigma(z),$$

then $p(z)$ is a polynomial and $p(\lambda') = 0$. Since $\lambda'_1, \lambda'_2, \dots, \lambda'_d$ are simple, $p(z) = \prod_{i=1}^d (\lambda'_i - z)$. Since the product of the roots is $p(0)$, the result follows. $\square$

Proof of Lemma 2. Assume the result holds for Z . By Lemma 1,

$$\begin{aligned} \det(S(Z + v \otimes v)S) &= \det(SZS + Sv \otimes Sv) \\ &= \det(SZS)(1 + Sv \cdot (SZS)^{-1}Sv) \\ &= \det(Z) \det(S)^2 (1 + v \cdot Z^{-1}v) = \det(Z + v \otimes v) \det(S)^2, \end{aligned}$$

hence the result holds for $Z + v \otimes v$. If $Z = \delta I$, the result holds. Since every $Z > 0$ is a sum of δI and a sum of matrices of the form $v \otimes v$, the result follows by induction over the number of terms in the sum. $\square$

Proof of Lemma 3. The first part is checked by differentiation. For the second part, clearly $h(\lambda) \leq c$ implies $\lambda \leq 1/\epsilon$. Since $\lambda \log \lambda \geq -1/e$ and $(e-1)e^c \geq ec$ for $c \geq 1$, $h(\lambda) \leq c$ and $c \geq 1$ implies $\lambda \geq (1-1/e)/c \geq \epsilon$. $\square$

Proof of Lemma 4. Since $Q^* > 0$, there is $S > 0$ with $Q^* = S^2$. Let $Z = S^{-1}QS^{-1}$ and let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Z . Then $\text{trace}(Q^{-1}Q^*) = \text{trace}(SQ^{-1}S) = \text{trace}(Z^{-1})$. By Lemmas 2 and 3,

$$\begin{aligned} 2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) &= \log \det Q - \log \det Q^* + \text{trace}(Q^{-1}Q^* - I) \\ &= \log \det(SZS) - \log \det(S)^2 + \text{trace}(Z^{-1} - I) \\ &= \log \det Z + \text{trace}(Z^{-1} - I) \\ &= \sum_{i=1}^d (h(\lambda_i) - 1) \geq 0, \end{aligned} \tag{16}$$

establishing $F_*(Q, Q^*) \geq F_*(Q^*, Q^*)$. If (16) is an equality, $Z = I$ hence $Q = Q^*$. Now suppose $F_*(Q, Q^*) < c$. Then

$$2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) \leq 2c - \log \det(2\pi Q^*) - \text{trace}(I) = \log(1/\epsilon) - 1.$$

Since the terms in the sum in (16) are nonnegative, $h(\lambda_i) - 1 \leq \log(1/\epsilon) - 1$ for $i = 1, 2, \dots, d$. By Lemma 3, $\epsilon \leq \lambda_i \leq 1/\epsilon$ for $i = 1, 2, \dots, d$. This implies $\epsilon I \leq Z \leq I/\epsilon$, hence $\epsilon Q^* \leq Q \leq Q^*/\epsilon$. $\square$

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